

Project Title

SentinelAI: A Multi-Agent Autonomous Self-Healing SOC Analyst for Predictive Threat Detection, Root Cause Investigation, and Intelligent Response

Idea Summary

SentinelAI is an **AI-powered security system that stays attached to a server 24/7** and continuously monitors it for attacks, malware, suspicious behavior, and policy violations. It works like an autonomous SOC analyst for backend systems.

The system is built to **understand server activity across multiple backend environments and programming ecosystems** such as Python, Node.js, Java, PHP, Go, .NET, Bash, and Linux/Windows server logs. Instead of depending on one specific stack, it uses **language-agnostic log parsing, pattern detection, and AI reasoning** to analyze threats regardless of the backend technology in use.

When a threat is detected, SentinelAI does not just raise an alert. It:

- identifies the threat,
- investigates the root cause,
- estimates severity,
- decides whether it is safe to auto-fix,
- applies safe remediation automatically for low and medium threats,
- and immediately emails the maintenance team with a full report.

For **critical or sensitive threats**, it does **not** blindly change the server. Instead, it generates a recommended fix, explains the risk, and alerts the team instantly so a human can approve the action. This makes the system both **intelligent and safe**.

How It Works

1. Continuous Monitoring

The agent is installed on the server and continuously watches:

- system logs
- process activity
- file changes
- authentication events

- network traffic
 - suspicious script execution
 - privilege escalation attempts
 - malware signatures
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2. Threat Detection

The AI detects anomalies such as:

- brute-force login attempts
 - malware execution
 - unusual outbound traffic
 - unauthorized file changes
 - suspicious cron jobs or persistence scripts
 - privilege escalation
 - data exfiltration attempts
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3. Investigation

Once something suspicious is found, the system investigates automatically:

- What happened?
 - Which user or process caused it?
 - Which file, service, or port was affected?
 - Is this a known attack pattern?
 - How severe is it?
 - Is it safe to fix automatically?
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4. Decision Engine

The system classifies the threat into three levels:

Low Risk

Automatic fix is applied immediately, then the team is notified.

Medium Risk

Automatic containment or repair is applied if safe, then a report is sent.

Critical Risk

No automatic destructive action is taken.
The system generates a fix recommendation and alerts the team instantly.

5. Email Alerting

Every incident is sent to the maintenance/security team through email with:

- threat type
 - severity
 - affected server
 - timestamp
 - root cause
 - action taken
 - recommended next step
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Example

Scenario:

A suspicious script tries to run on the server and starts creating outbound connections to unknown IPs.

SentinelAI response:

- detects abnormal process behavior,
- identifies it as possible malware,
- checks whether the process is isolated,
- kills it automatically if safe,
- quarantines the suspicious file,
- updates the incident log,
- and sends an email to the server maintenance team.

Critical case:

If the system detects possible ransomware-like file encryption or root-level compromise, it will:

- stop automatic destructive actions,
 - generate the safest fix plan,
 - isolate the incident,
 - and immediately notify the team for approval.
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Tech Stack

Frontend

- Next.js
- TypeScript
- Tailwind CSS
- ShadCN UI
- Recharts
- Socket.IO Client

Backend

- FastAPI
- Python
- Celery
- Redis
- WebSockets

AI / Agent Layer

- LangGraph
- LangChain
- Google Gemini / OpenAI API
- Hugging Face Transformers
- PyTorch
- Scikit-learn

Detection & Security

- Sigma Rules
- YARA Rules
- MITRE ATT&CK mapping
- Wazuh
- OSQuery
- Auditd
- Sysmon

Data & Storage

- PostgreSQL
- Elasticsearch
- MongoDB
- Redis

Deployment & Infra

- Docker
- Kubernetes
- Nginx
- GitHub Actions
- AWS / Azure / On-prem server

Monitoring

- Prometheus
 - Grafana
 - OpenTelemetry
 - Loki
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One-Line Final Pitch

SentinelAI is an autonomous self-healing SOC agent that continuously protects servers, detects threats early, investigates incidents intelligently, auto-fixes safe problems, and escalates critical threats to humans before damage occurs.

Why This Idea Is Strong

- practical for real server security
- works continuously without human supervision
- supports multiple backend ecosystems
- includes autonomous response
- still keeps human approval for critical cases
- combines AI, cybersecurity, and self-healing infrastructure
- sounds research-worthy and project-worthy